

I Built a Kubernetes Hot Rod

...and I Learned More Than I Ever Could Driving a Tesla



Speaker: Philip Brown (koob.foo)

Production Kubernetes distributions are like Teslas; smooth, polished, they get you where you need to go, and in style. But if you want to truly understand how Kubernetes works, sometimes you need to strip it down to the chassis and get your hands dirty with every component.

I built **koob OS**, a 89MB Kubernetes distro from scratch, replacing Systemd with a custom Go-based init system, eliminating GNU tools and package managers, and implementing UEFI Secure Boot with a Unified Kernel Image. This talk shares my journey: implementing PID 1 in Go, making containerd work without standard userspace, and discovering what Kubernetes actually needs to run.

You'll see a live build-to-boot demo and learn about zero-trust boot environments, immutable infrastructure patterns, and the minimal requirements for running Kubernetes. This isn't a production recommendation; it's a mechanics class that taught me more about Kubernetes internals than years of driving production clusters ever could.